Recap: Lecture 29

In Lecture 29, we

- Introduced a FT table
- Introduced convolution and shift theorems for FT

$$\mathscr{F}\{f * g(x)\} = \widehat{f}(k)\,\widehat{g}(k), \quad \mathscr{F}\{f(x-y)\} = \mathrm{e}^{-iky}\,\widehat{f}(k) \qquad (y \in \mathbb{R}),$$

$$\mathscr{F}\{\mathrm{e}^{iqx} f(x)\} = \widehat{f}(k-q) \qquad (q \in \mathbb{R})$$

- Learnt rules to help apply FTs to ODEs

$$\mathscr{F}\{f^{(n)}(x)\} = (ik)^n \widehat{f}, \quad \mathscr{F}\{x^n f(x)\} = i^n \frac{\mathrm{d}^n \widehat{f}}{\mathrm{d}k^n}$$

Lecture 30

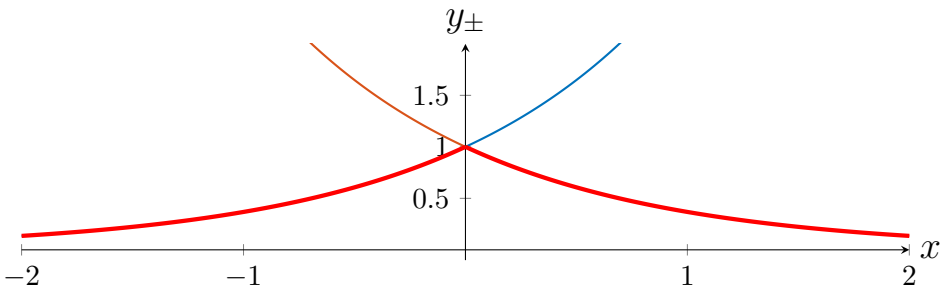
In Lecture 30, we will:

1. Apply the FT to the Airy ODE
2. Apply the FT to a PDE (1D heat equation)

Exercise 6.9 revisited

$$y''(x) - y(x) = -2\,\delta(x) \quad \longrightarrow \quad \text{FT} \quad \longrightarrow \quad y(x) = -\frac{1}{2}\mathrm{e}^{-|x|}$$

$$y''(x) - y(x) = 0 \quad \Rightarrow \quad y(x) = C_+\, y_+(x) + C_-\, y_-(x) \quad \text{where} \quad y_\pm(x) = \mathrm{e}^{\pm x}$$



Recap (Topic 2): Airy's equation $y''(x) - x\,y(x) = 0$

Has no solutions in terms of elementary functions

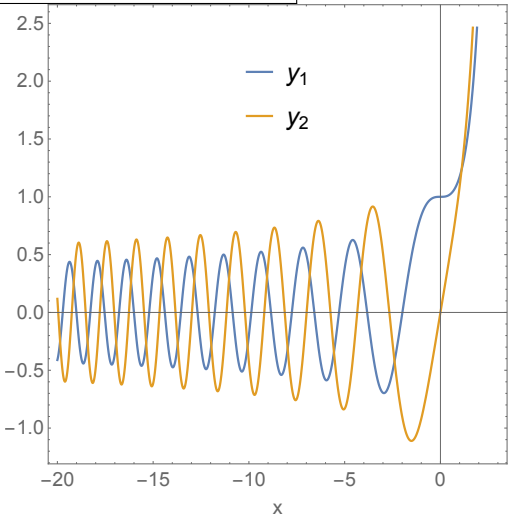
Two independent series solutions:

$$y_1(x) = 1 + \sum_{k=1}^{\infty} \frac{x^{3k}}{2 \cdot 3 \cdot 5 \cdot 6 \cdots (3k-1) \cdot 3k}$$

$$y_2(x) = x + \sum_{k=1}^{\infty} \frac{x^{3k+1}}{3 \cdot 4 \cdot 6 \cdot 7 \cdots 3k \cdot (3k+1)}$$

The general solution of Airy's eqn.

$$y(x) = C_1\, y_1(x) + C_2\, y_2(x)$$



Recap (Topic 2): Airy functions

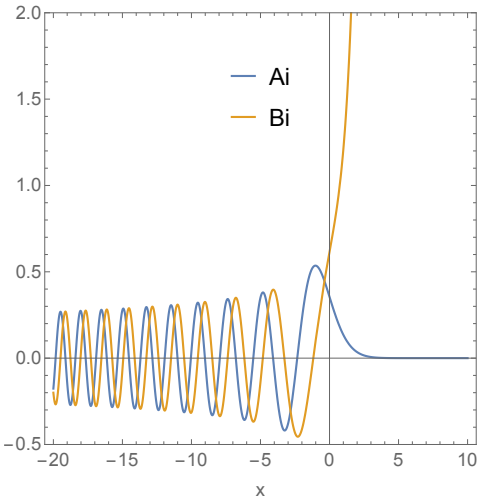
The *Airy functions* are

$$\mathrm{Ai}(x) = \alpha\big[y_1(x) - \beta y_2(x)\big]$$

$$\mathrm{Bi}(x) = \sqrt{3}\alpha\big[y_1(x) + \beta y_2(x)\big]$$

where $\alpha \approx 0.355$ and $\beta \approx 0.729$.

Note $\mathrm{Ai}(x) \to 0$ as $x \to \infty$



Exercise 6.10: Use FTs to solve Airy's equation $y''(x) - xy(x) = 0$

Applying the Fourier transform to the ODE we find

$$-k^2\widehat{y} - i\widehat{y}' = 0$$

This differential equation has the general solution

$$\widehat{y}(k) = Ae^{\frac{i}{3}k^3}$$

Choosing $A = 1$ and performing an inverse Fourier transform results in

$$y(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{\frac{i}{3}k^3 + ikx}\, dx$$

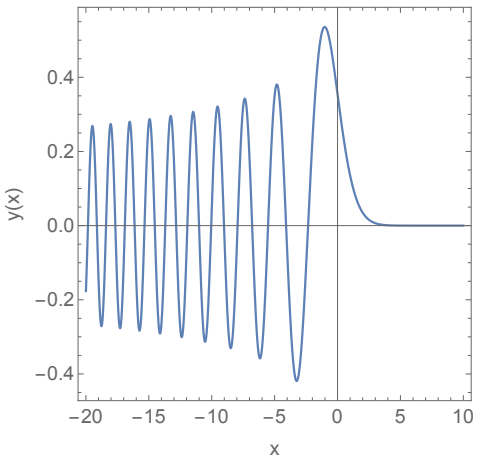
This is precisely the Airy function $\mathrm{Ai}(x)$ (at least up to a scalar multiple)

Why? Using Fourier transforms requires integrability at $\pm\infty$!
However, $\mathrm{Bi}(x)$ is not integrable at ∞

We have found Ai in integral form

$$\mathrm{Ai}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}\,(k^3/3 + k\,x)}\ \mathrm{d}k$$

More on this function in MAST90067 Advanced Methods: Transforms



Transforms applied to PDEs

- We've already seen that Laplace transforms and Fourier transforms turn ODEs into algebraic equations
- Transforms can also be used to turn PDEs into ODEs
- Let $u(x,t)$ be defined for $x \in \mathbb{R}$ and $t > 0$.
  - Let $\widehat{u}(k,t) = \mathscr{F}\{u(x,t)\}$
  - $\mathscr{F}\left(\dfrac{\partial^n u}{\partial x^n}\right) = (ik)^n\,\widehat{u}(k,t)$, etc.

Exercise 6.11: 1D Heat Equation

$$\dot{u}t = \alpha u'' \quad \text{where} \quad \alpha > 0, \quad x \in \mathbb{R}, \quad t > 0,$$

subject to the initial condition $u(x,0) = f(x)$ and $u(x,t) \to 0$ for $x \to \pm\infty$.

The boundary conditions for $u(x,t)$ mean that we can use the Fourier transform in x. Define

$$\widehat{u}(k,t) = \mathscr{F}\{u(x,t)\} = \int_{-\infty}^{\infty} u(x,t)e^{-ikx}\,dx$$

Taking the derivative with respect to t we find

$$\partial_t \widehat{u}(k,t) = \mathscr{F}\{\dot{u}(x,t)\} = \mathscr{F}\{\alpha u''\} = -\alpha k^2 \widehat{u}(k,t)$$

This is an ODE with general solution

$$\widehat{u}(k,t) = A(k)e^{-\alpha k^2 t}$$

Note that the coefficient $A(k)$ may depend on k.

It remains to impose the initial condition. We could transform back to x-space and impose it directly. Instead, we use

$$\widehat{u}(k,0) = \int_{-\infty}^{\infty} u(x,0)e^{-ikx}\,dx = \int_{-\infty}^{\infty} f(x)e^{-ikx}\,dx = \widehat{f}(k)$$

This permits to identify $A(k) = \widehat{f}(k)$.

The final solution is thus given by

$$u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(k)e^{-\alpha k^2 t}e^{ikx}\,dk$$

Find the particular solution for $f(x) = \delta(x)$.

For this initial condition we find

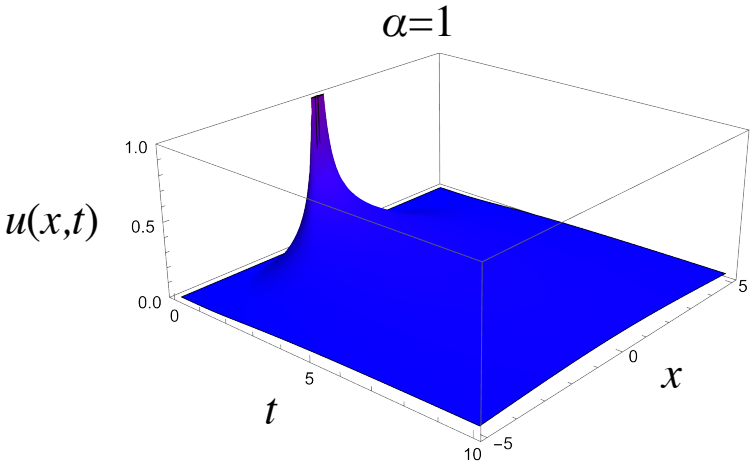
$$\widehat{f}(k) = 1$$

and hence

$$u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\alpha k^2 t} e^{ikx}\, dk = \frac{1}{2\sqrt{\pi a t}}\, e^{-\frac{x^2}{4at}}$$

This type of solution is called a "heat kernel".

Apart from applications it plays a crucial role in pure mathematics, e.g. in the proof of the celebrated Atiyah-Singer index theorem which links the areas of analysis and topology


α=1
u(x,t)
1.0
0.5
0.0
0
5
10
t
−5
0
5
x